



StepOut: Frictionless Route-Based Exercise App

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Introduction & Goals:

StepOut helps people fit exercise into their busy days. Instead of requiring planning and an abundance of time all you need is a drive to get better. Users can insert a time or distance and a random route is generated for them.

Goals:



Make movement easy to start

Remove barrier and decision fatigue



Eliminate Planning

Generate routes automatically



Provide usable, time aware routes

Routes match time or distance you give

Core Features:



Route Generation

One-way, Out-and-back, loop



Pacing System

Has walking, jogging, and running



User Specific Routes

Times are based off of user's speeds



Route Saving with Core Data

Save and reuse previous routes



Live and Accurate Tracking

Allows to see progress in real time



Voice directions, haptics, and Widgets

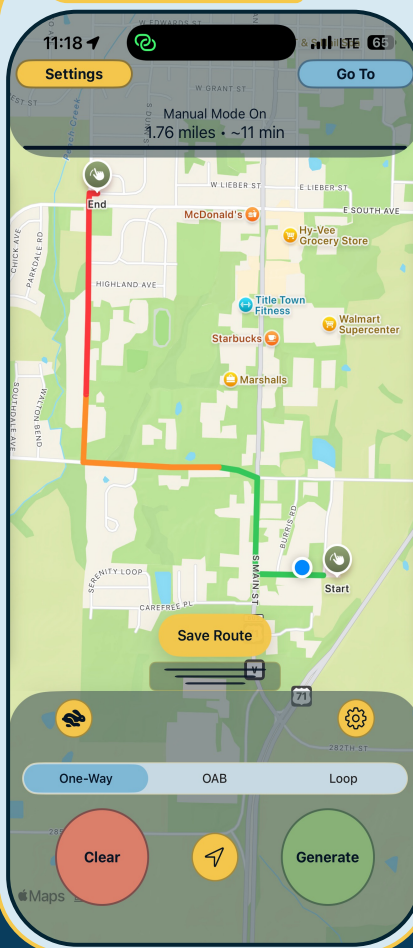
Stay guided and motivated



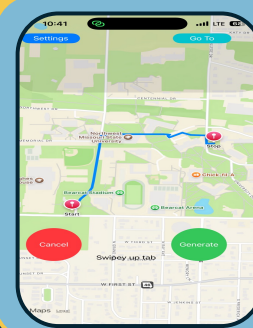
Tutorial System

Allows anyone to use the app quickly

Current Interface



Early Version of StepOut



- UI is extremely basic.
- Only has manual pin placement
- Does not track users' location.
- Only 1 color of route.
- No set colors or general theme.
- Constraints are not correct.

What I Found

- UI clarity is just as important as system features.
- Connecting simple systems together quickly becomes complicated.
- UI polish takes a long time and is needed for a functional tool.
- User insight is important but not the final say
- Longer projects are significantly harder but more satisfying than shorter projects.
- Solo development showed me the value of specialized roles.

Challenges & Solutions

- **Route Distance Inconsistency**
Solution: Retry System + Radius Adjustments
- **UI complexity & Discoverability**
Solution: Side panels & Tutorial
- **Random Route Generation Failing**
Solution: Fallback logic & improved validation
- **Accurate Times with Pace Changes**
Solution: Use pace-based segment calculations

Future Additions

- Apple watch functionality
- Challenge/Achievements system
- Calorie usage estimator
- Ability to factor in weather
- FAQ/Way to contact me

Analysis & Dev Process

- Started with a simple MapKit route prototype.
- Added time-based routing, pace segments, and random generation.
- Tested route accuracy and adjusted generation logic.
- Improved usability through feedback and a spotlight tutorial.
- Used A.I. for refining animations and difficult debugging.



Choose time or distance

User gives how far or long they want the route to be.



Route Type + Pace selection

Pick One-way, OAB, or loop. Then set pace pattern and percent.



Generate Route

App makes route automatically using Mapkit.



Adjust for Accuracy

Route is checked against target and adjusted if needed



Guided Walk

User gets progress, pace, voice, haptics, and Live Activity updates

Acknowledgments:

Thanks to Dr. Hoot and NWMSU for the guidance and chance to build this app.

Additional Resources:

Apple Developer Documentation (Mapkit, Core Data, Widgets, Live activities, etc.)
Swift and Xcode development tools
Figma A.I. Tool, ChatGPT, and Claude



GitHub Repository:

[Click here to visit repo](#)